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Procedia Computer Science 246 (2024) 159–168

Procedia Computer Science

www.elsevier.com/locate/procedia

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

VR running system using redirected walking technique

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Abstract

This study presents a novel virtual reality (VR) running system designed to allow users to simulate the experience of running outdoors while exercising indoors. This is achieved through the use of a head-mounted display (HMD), providing users with a immersive running experience at their convenience. The proposed system has three main functions: a redirected walking technique to map the direction of movement between real and virtual spaces, a virtual nose and Meta Quest 2 Guardian function to prevent VR sickness, and 3D CG models using photogrammetric technology to enhance the sensation of running outdoors. The results of an experimental evaluation of 17 participants demonstrated that approximately half felt the sensation of running; however more than 60% felt VR sickness while using the system. Subsequent exploratory experiments suggested that the Space Sense function of Meta Quest 2 could improve VR sickness. Despite numerous studies on walking with HMDs, to our knowledge, this is the first study on running in relatively small spaces, and the findings offer insights into future improvements and developments in this field.

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Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: redirected walking; sense of real running ; VR sickness; virtual nose; photogrammetry

1. Introduction

Running has been popular among many people as a health-promoting exercise because of its accessibility, requiring no special equipment to begin. Lee et al. reported that people with a running habit have a 30% lower risk of death, a 45% lower risk of dying from heart disease, and potentially enjoy an extended life expectancy of approximately three years compared to non-runners [1]. Although running is generally an outdoor exercise, it is affected by weather and temperature, leading several people to use treadmills in fitness gymnasiums. Treadmill exercise is a type of aerobic exercise that requires a minimum of 20 min of continuous exercise. People highly interested in improving their physical condition and weight loss are willing to perform these exercises. However, running for 20 min is painful for those with no exercise habits, and it is not easy to continue the exercise.

Numerous support systems that use virtual reality (VR) technology have been developed to motivate people without exercise habits to exercise in the real world. Octonic has created a VR application, Octonic VR, that is compatible

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10.1016/j.procs.2024.09.238

with nearly all treadmills on the market [2] (Figure 1a). Users can select various running locations, such as trekking courses and parks. The user can start and stop running and adjust the speed using a control panel that appears in the VR space, allowing the user to exercise while wearing a head-mounted display (HMD). Virtuix has developed the Virturix Omni treadmill designed for virtual reality space [3] (Figure 1b). In general, treadmill running is only in the forward direction, however, the Virturix Omni can move in all directions. The system can be used for activities other than running, such as games and sports.

While these systems provide a variety of activities to encourage individuals without exercise habits to engage in physical activity, they require the purchase of a treadmill and a place for setup and use. Virturix Omni users are required to wear special shoes and belts. It is difficult for them to start running quickly in their rooms if it takes time and effort to prepare. Therefore, we propose a VR running system that does not require any special equipment other than an HMD. This design enables even those without established exercise routines to commence exercising promptly as need arises.

The main contributions of this study are as follows:

- A redirected walking technique that converts movement in a small indoor space into movement on a running course in a virtual space and a VR sickness countermeasure.
- A 3D CG model of a running course using photogrammetric technology to enhance immersion in virtual space.
- Short-term evaluation of the proposed system by 17 participants using a running course replicating the area around a university gymnasium and long-term evaluation by one participant.

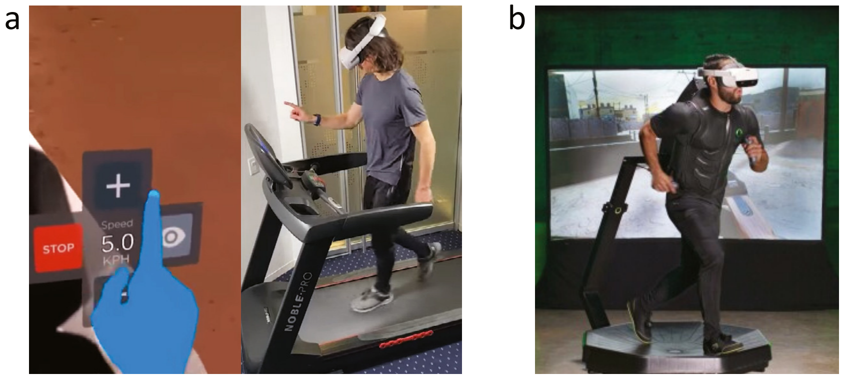

Fig. 1. (a) Octonic VR; (b) Virturix Omni.

2. Design and Implementation

2.1. Overall view of the system

The proposed system runs on a standalone HMD. No treadmill or controller is used. When the user starts the application with the HMD, they see a first-person view from the starting point of an outdoor running course (Figure 2). The user runs in a circle around the open space in the room. The lower limit of the circle diameter was set to 3 m, assuming a typical Japanese house. The system has four main functions, as described in the next section. The redirected walking technique is a software algorithm that converts a user's motion vector in real space into one in a virtual space. The virtual nose and the Guardian function are 3D CG objects, both of which are used to countermeasure VR sickness. Photogrammetry is a technique for creating a realistic 3D CG object that allows users to experience high immersion in a virtual space.

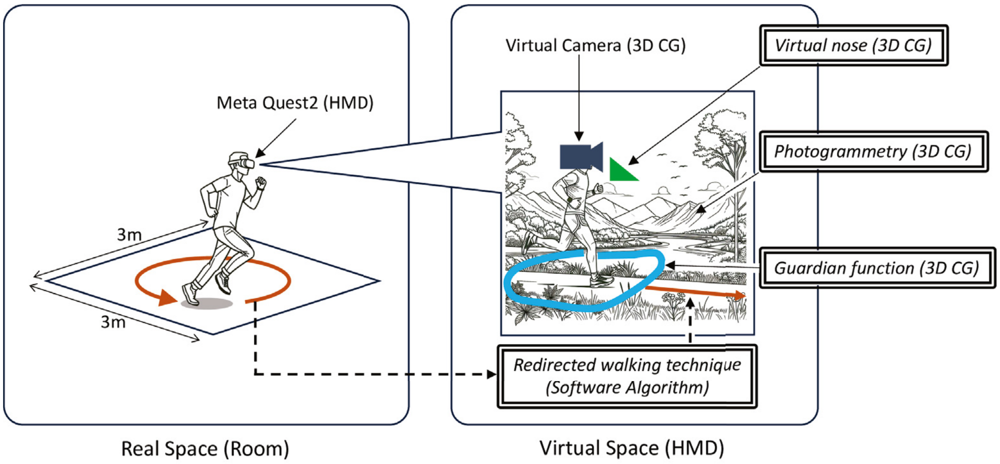

Fig. 2. Overall view of the system.

2.2. Redirected walking technique

In general VR applications, the movements of the user's body and head are reflected in the movements of the virtual camera in virtual space in real-time. If the user moves forward in real space, the viewpoint in virtual space also moves forward. In contrast, in the proposed system, the directions of the user's movements in the real and virtual space are different. In real space, the user runs in a circle around the room; however, in virtual space, the user runs almost straight ahead of the running course designed in advance. To map this real space movement to a virtual space movement, a redirected walking technique (RDW) is used in this study [4].

Although many studies have been conducted on RDW, one common characteristic is using the human illusion of movement in a restricted indoor space to map movement in a virtual space to move in a real space. In this approach, the distance the user moves in real space is obtained, and the movement in the virtual space is calculated by multiplying the distance moved by the movement vector at each position of the running course in virtual space (Figure 3). Equation 1 is used for this calculation, where $P_r(t)$ represents the two-dimensional position vector in real space at time t. This study used Meta Quest 2 from Meta as the HMD [5]. This HMD is equipped with a gyro sensor so the user's relative position on a plane in real space can be obtained in real time. This value is set to $P_r(t)$, where $P_v(t)$ is a two-dimensional position vector in the virtual space. The distance moved in real space is calculated by $|P_r(t) - P_r(t-1)|$ and multiplied by the movement vector $V_v(P_r(t-1))$ at the position at time $t-1$ to calculate the position vector $P_v(t)$ in virtual space at time t. α is the adjustment factor for the amount moved in real and virtual space.

$$P_v(t) = \alpha \cdot |P_r(t) - P_r(t-1)| \cdot V_v(P_v(t-1)) + P_v(t-1) \tag{1}$$

2.3. Countermeasures against VR sickness

One of the problems with virtual reality experiences using HMDs is VR sickness, which can be caused by sensory conflicts when visual information does not match dynamic vestibular input [6]. Countermeasures against VR sickness are vital because the motion of the proposed system is such a motion that it causes precisely this kind of sensory discrepancy. The following two measures were taken in this study based on preliminary experiments. A description of the experiment is omitted to save space.

2.3.1. Virtual nose

In reality, our field of vision includes our nose, eyeglass frames, and other objects in our field of vision. Placing this fixed object in front of the camera in virtual space reduces VR sickness [7]. David et al. reported that when participants were allowed to use a VR application without telling them whether they had a nose, they could use the VR application

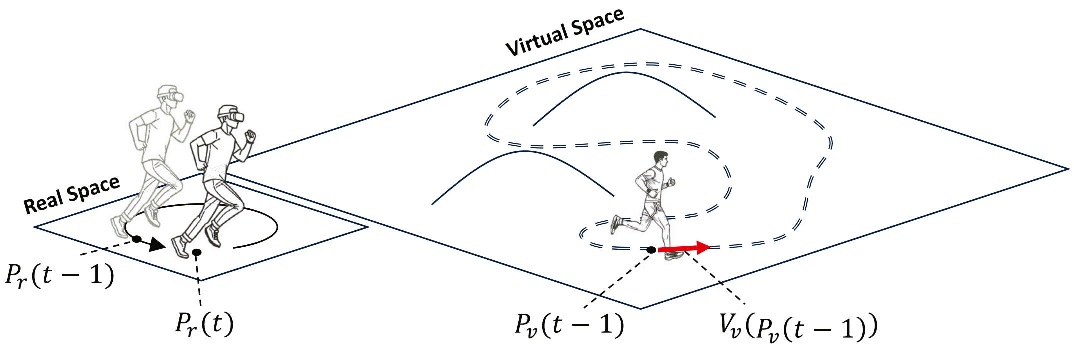

Fig. 3. Diagram of the correspondence between the amount of movement and the direction of movement in real and virtual space.

longer if they placed their nose in the virtual space. In the proposed system, a 3D CG object of the nose was placed under a virtual camera in virtual space (Figure 5b).

2.3.2. Guardian function

A standard feature of Meta Quest 2 is that it allows users to specify the range in which they can move in real space to ensure safe use of the HMD [8]. The range can be determined by drawing a boundary line in real space while wearing the HMD (Figure 4a). When using VR applications, this boundary line is always visible (Figure 4b). By setting a boundary line in the virtual space, users can run while being aware of the actual space, which reduces VR sickness.

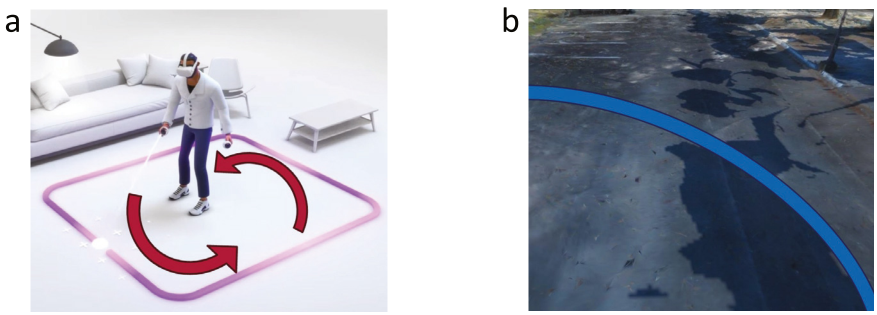

Fig. 4. (a) This illustration shows a user using Meta Quest 2's Guardian function to set the boundaries of the playable area. These boundaries are mapped in real space; thus, they remain even when the user moves. (b) An example of the boundary line displayed on the running course in virtual space.

2.4. Photogrammetry

To bring the virtual space closer to the actual space, we created a 3D CG model of an actual running course by scanning it in 3D. There are various methods for 3D scanning [9]. This study used photogrammetry and its software, 3DF Zephyr [10]. Photogrammetry is a technique used to photograph subjects from various angles and integrate photographs to create a 3D CG model [11]. A running course was set up around the gymnasium at the Kindai University (300 m per lap). A 3D CG model was created by taking 10,000 photographs of the course using a single-lens reflex camera (EOS Kiss X90). Figure 5a illustrates the entire 3D CG model of the running course, and Figure 5b illustrates an example of the image projected on the HMD.

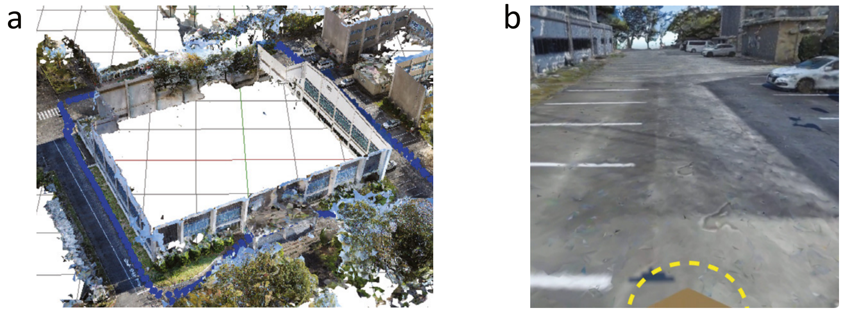

Fig. 5. (a) Overall view of the 3D CG model of the running course created with the photogrammetry software 3DF Zephyr based on photographs taken around the 300 m perimeter of the gymnasium. The blue dots represent running courses. (b) From the user's perspective, running course and virtual nose object (yellow dashed line).

3. Evaluation experiment

3.1. Experiment 1 (short-term evaluation)

This experiment aimed to verify whether the proposed system feels like running outdoors. This sensation is called the sense of real running (SRR) in this study. We also verified how strongly the participants experienced VR sickness while using the proposed system. Seventeen participants (13 males and four females), aged between 19 and 25, were selected. The participants ran in a circle with a radius of 1.5 m in real space while wearing the HMD. Participants determined their running speed. The participants continued running until they completed one lap (300 m) of the running course projected on the HMD. Immediately after the experiment, participants were asked to complete a questionnaire. Participants answered Q1 through Q3 with a score on a 5-point scale for the intensity of the sensation and the reason for it, and Q4 and Q5 with a Yes/No response. As the participants were all students at Kindai University, they had experienced walking the running course displayed in the virtual space, which contributed to the accuracy of the answers to Q1.

Q1: To what extent did you feel the sensation of running around the gymnasium in real space?
Q2: To what extent did you feel satisfied with your exercise program?
Q3: To what extent did you experience VR sickness?
Q4: Do you have exercise habits daily?
Q5: Do you experience VR sickness?

3.2. Results of experiment 1

All participants completed the VR course. The average running duration was 252 s (SD=38). Each course lap was approximately 300 m long, meaning the average running speed was 4.4 km/h (SD=0.7), which indicates that running is similar to light jogging. Figure 6 illustrates the distribution of scores for the responses to Q1 through Q3. Q2 showed that many participants responded with higher scores than Q1 and Q3. To analyze the cause of these responses, we divided the participants into two groups according to their exercise habits in Q4 and VR experience in Q5. We conducted a t-test at the 5% level of significance for the responses of Q1 to Q3. The results are shown in Figures 7 and 8. Significant differences were found only for VR experience and SRR, which indicates that those with no VR experience had a lower SRR and satisfaction.

3.3. Experiment 2 (long-term evaluation)

In this experiment, the proposed system was used repeatedly in an exploratory process to identify and rectify issues, ensuring its viability for prolonged use. The experimental location and methodology were the same as in experiment 1. The experiment was started at 18:00 every day for five consecutive days. The goal was to complete three laps of a

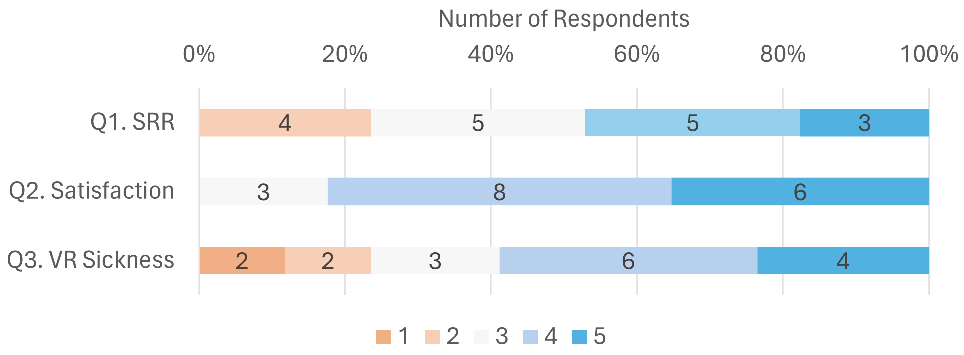

Fig. 6. Distribution of the number of responses to the questionnaire regarding the intensity of sensations in Q1 to Q3 immediately after using the proposed system.

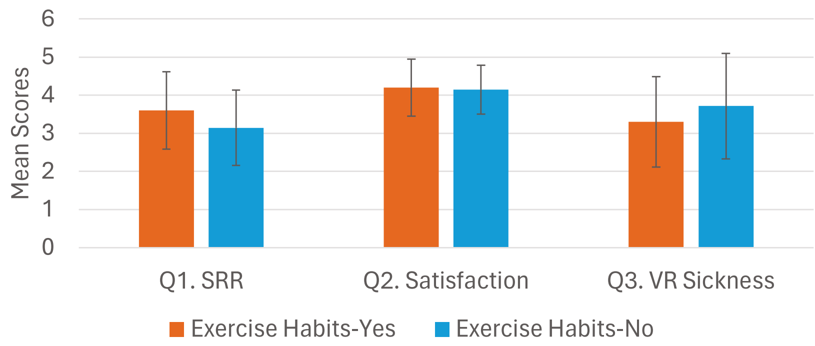

Fig. 7. The mean scores of Q1 to Q3 were used to compare the two groups of experimental participants, with and without exercise habits.

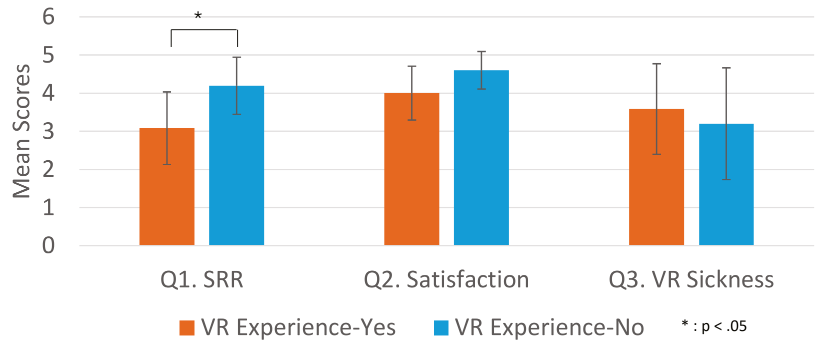

Fig. 8. The mean scores of the Q1 to Q3 questionnaires were used to compare the two groups of experimental participants with and without VR experience.

running course in a virtual space, and the time of use, satisfaction, VR sickness, and degree of eyestrain were recorded on a 5-point scale after the exercise. The subject was an author (male, 24 years old). Note that the subject was a member of a track and field team at a high school.

3.4. Results of experiment 2

The experimental results are presented in Figure 9. This shows the evaluation of the system and the improvement process recorded from Days 1 to 5. Although the boundary assured safety, the running speed could not be increased because of the fear of being unable to see real space. Therefore, the participant did not feel as satisfied as he did after the experiment. Repeating the experiment on the second and third days allowed the participant to become accustomed to the proposed system. The participant was able to look straight ahead. As he did, the surrounding scenery entered his eyes, and he experienced VR sickness. Therefore, the patient could not complete the three laps, and the time spent using the system decreased on the second and third days. As he thought it would be difficult to get used to it even if he continued the experiment, he activated the Space Sense function of Meta Quest 2 on the fourth day. The Space Sense function displays an outline of the obstacles detected by an external camera of Meta Quest 2. By setting up a table with a radius of 1 m in the center of the running room, the outline of the table is always visible. Running with this image stabilized running, and on the fourth and fifth days, VR sickness weakened, allowing him to complete the running course.

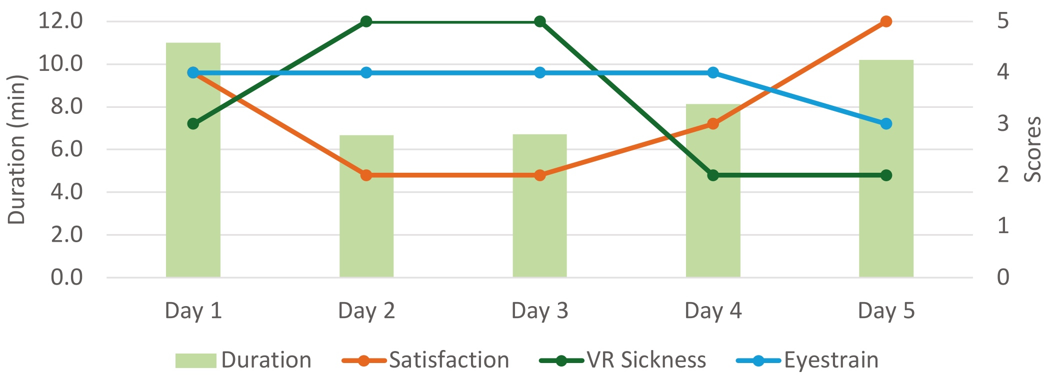

Fig. 9. The duration spent by one participant using the proposed system over a 5-day period (bar graph) and the score of satisfaction and intensity of the sensation of VR liking and eyestrain after using the system were answered on a 5-point scale (line graph).

4. Discussion

4.1. SRR

Twenty-three percent (4/17) responded two or less to Q1, and 47% (8/17) responded four or more. Approximately half of the respondents reported feeling like an SRR. The reasons for the high scores were that *“The images in the virtual space changed in real-time according to the movements in the real space, so it felt close to actual running,”* and *“The changing scenery reminded me of the actual running.”* The RDW and photogrammetry of the proposed system may have contributed to SRR. The t-test showed a significant difference between the VR experience and SRR, but those without VR experience felt a higher sense of running. This result contradicts our hypothesis. The proposed system exhibits different movements in the real and virtual spaces. Because this type of motion is not common in VR applications, we believe that the difference between the expectation based on past VR experience and the motion of the proposed system may have led to a low score.

The main reasons for the low score were: *“I felt anxious about colliding with objects because I could not see the real space when I was running through the boundary line,”* and *“I couldn’t feel the change of scenery because my eyes were drawn to the boundary line.”* As one of the countermeasures against VR sickness, boundary lines were displayed to be always visible, but this reduced the sense of immersion in the virtual space. To counter this problem, measures such as selectively displaying the boundary line when the user wants to see it should be considered.

4.2. Satisfaction

The percentage of participants who answered four or more to Q2 was 82% (14/17). Most participants indicated that they felt satisfied after the exercise. The reasons given for their responses included *"The physical movements were the same as actual running and I felt similar fatigue after the exercise,"* and *"I circled around a gymnasium that I had actually seen, so I felt like I was running in that place."* The t-test results demonstrated no significant difference in satisfaction between the participants with and without exercise habits. The t-test results showed that satisfaction after exercise was obtained regardless of the usual exercise habits. There was also a response of *"I felt satisfied after running because I don't usually exercise."* The fact that even those who do not have exercise habits can experience exercise similar to running suggests that the proposed system is effective in allowing people to exercise without having to go out when they want to exercise.

4.3. VR Sickness

The percentage of participants who answered two or less to Q3 was 23% (4/17), and 59% (10/17) answered four or more. More than half of the participants experienced intense VR sickness. To investigate the cause of VR sickness, we examined the correlation between running speed and VR sickness. The results showed a weak negative correlation (r=-0.24, $R2$=0.06). Specifically, the higher the running speed, the lower is the VR sickness. It is suggested that taking measures to increase the running speed may reduce VR sickness. We believe that the results of Experiment 2 supported this hypothesis. Specifically, equilibrium was maintained because the user could see objects around him using the Space Sense function. Consequently, participants were able to run safely. We believe that the running speed increased because the participant could see the objects around him using the Space Sense function.

5. Related work

5.1. RDW

In addition to RDW, research on mapping real and virtual space motions has included repositioning systems and proxy gestures. A repositioning system enables a walking motion at a fixed position by inhibiting the user's forward movement. The most common method is using treadmill-type equipment [2][3][12]. Because the motion in real and virtual spaces is consistent, VR sickness is less common; therefore, the cost of purchasing large equipment and a place to install it is necessary. Proxy gestures are a method of moving in a virtual space by moving body parts such as arms and legs. Liwei et al. proposed a system that moves within a virtual space by wearing shoes with attached sensors and stepping on them [13]. The system can be used simply by wearing a small device and has the advantage that it can be operated with less fatigue by stepping on one's feet while sitting down. However, satisfying the desire to perform running exercises would be difficult because they are different from actual running.

5.2. VR Sickness

The proposed system employs a virtual nose to counter the VR sickness. Park et al. proposed a method to dynamically extract the least moving content region and used it as a rest frame [14]. They reported a 37% reduction in VR sickness compared to a fixed FOV [15]. Milivoj et al. used a full-body controller called C-Infinity, which significantly reduces VR sickness by sending a signal to the brain region that perceives movement when a button for the movement operation is pressed, thereby making the user perceive that they are moving [16]. Because HMD wearers use both systems in a seated position, their applicability to the system proposed in this paper is unknown. However, evaluation experiments have demonstrated that the VR sickness countermeasures of the proposed system are insufficient, and we aimed to explore the possibility of improvement based on existing studies.

5.3. Exergame

The system proposed in this study aims to motivate people to exercise by making it similar to natural running. However, there exists a related research field known as Exergame, which employs methods to motivate individuals

[17]. Ioannou et al. used a virtual performance augmentation system to give users the illusion of greater ability than they have called virtual performance augmentation (VPA), which gives the user the illusion of higher actual ability. Virtual reinforcement of running and jumping has also been found to increase intrinsic motivation, perceived ability, and flow and may increase motivation for physical activity in general [18]. All of these factors are expected to promote motivation for exercise and continuation of exercise, which may improve athletic performance.

6. Limitations

The participants of the experiment were college students. If other age groups use the proposed system, actual running sensation and VR sickness may differ. In particular, VR sickness tends to be felt more strongly in older age groups [6]. Although the virtual nose and boundary line display using the guardian function were employed as VR sickness countermeasures, the degree to which they were gazed at was not obtained in the experiment; therefore, the relationship between the countermeasures employed in this study and the results of this experiment cannot be guaranteed. In addition, the simulator sickness questionnaire (SSQ) was used to evaluate VR sickness, and the participants were asked to indicate the intensity of VR sickness on a 5-point scale to align it with other question items, which may differ from the evaluation when the SSQ is used.

7. Conclusion

In this study, we proposed a system that allows users to feel like they are running outdoors by simply wearing an HMD, regardless of the weather or temperature. The system aimed to enable those with no exercise habits to continue exercising by synchronizing the scenery in the virtual space with indoor running exercise. In the evaluation experiment, approximately half of the subjects felt a sense of actual running and satisfaction, but no satisfactory results were obtained for VR sickness. The higher the running speed, the lower the VR sickness. It has been suggested that measures to increase the running speed can reduce VR sickness.

To our knowledge, this is the first study on running in a stenosis-free space. In the future, we will analyze the conditions that cause VR sickness in the proposed system and apply the VR sickness countermeasures listed in Section 5 to the proposed system to reduce VR sickness. In addition, although the proposed system is easy to start as soon as the HMD is put on, we believe that a system that provides entertainment and rewards, such as those shown in exergames, is necessary for continuous use. In the future, we intend to improve the system by combining these elements.

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